Longitudinal changes of manganese-dependent superoxide dismutase and other indexes of manganese and iron status in women.

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The effect of dietary factors on manganese-dependent superoxide dismutase (MnSOD) activity in humans has not been studied. We longitudinally evaluated changes in MnSOD activity and other indices of manganese and iron status in 47 women during a 124-d supplementation study. Subjects received one of four treatments: placebo, 60 mg iron, 15 mg manganese, or both mineral supplements daily. Manganese supplementation resulted in significant increases in lymphocyte MnSOD activity and serum manganese concentrations from baseline values but no changes in urinary manganese excretion or in any indices of iron status. Oral contraceptive use and the stage of the menstrual cycle did not confound the use of lymphocyte MnSOD activity or serum manganese to monitor manganese status, but fat intake affected both indices. This work demonstrated that lymphocyte MnSOD activity can be used with serum manganese concentrations to monitor manganese exposure in humans.